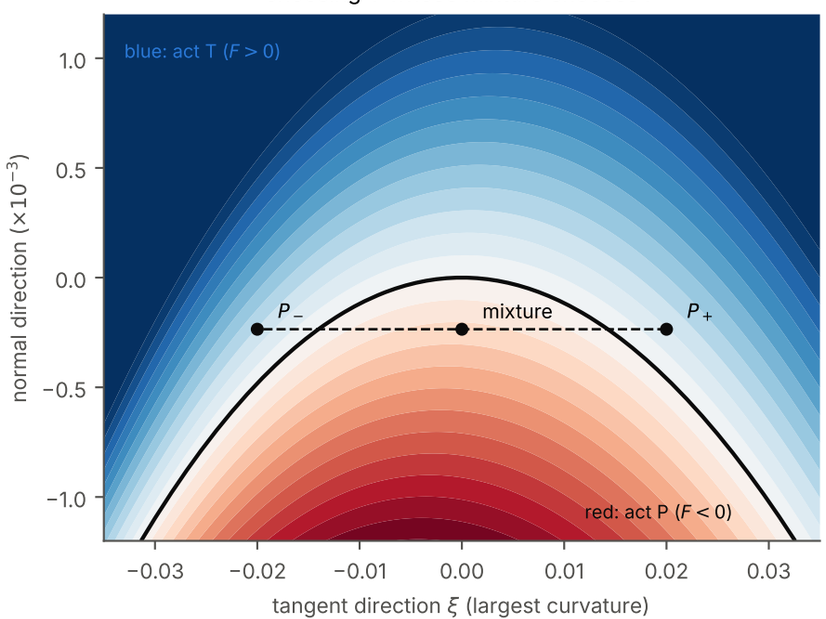
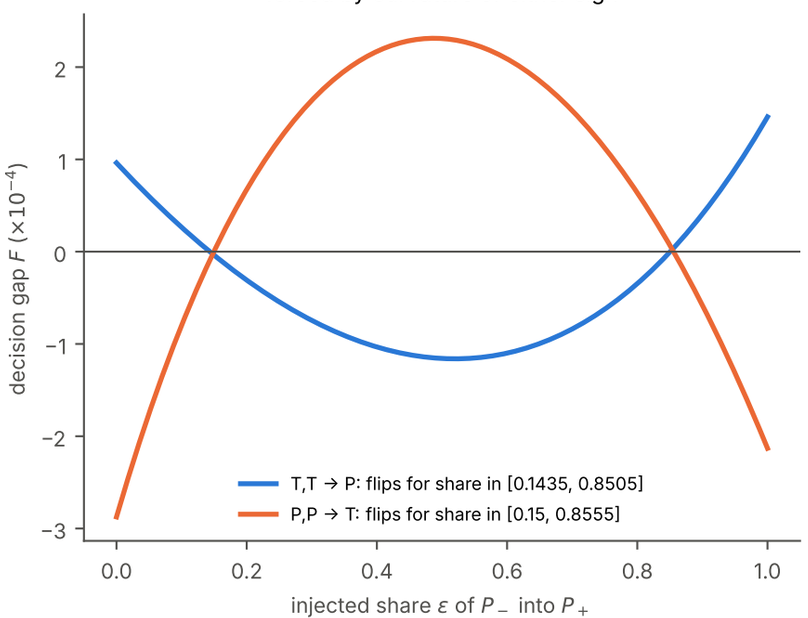
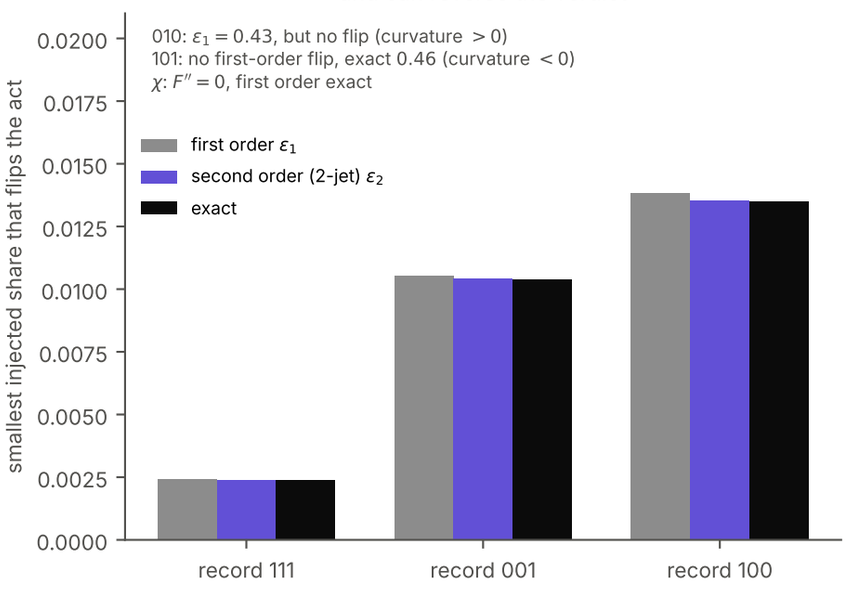

(a) the post-probe boundary curves: two laws
choosing T whose mixture chooses P
blue: act T ($F > 0$)
red: act P ($F < 0$)
P_-
mixture
P_+
normal direction ($\times 10^{-3}$)
tangent direction ξ (largest curvature)
1.0
0.5
0.0
−0.5
−1.0
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03


(b) pooling paradox: impossible for affine F,
forced by curvature of either sign
decision gap F ($\times 10^{-4}$)
injected share ε of P_- into P_+
T,T → P: flips for share in [0.1435, 0.8505]
P,P → T: flips for share in [0.15, 0.8555]
2
1
0
−1
−2
−3
0.0
0.2
0.4
0.6
0.8
1.0


(c) vertex attacks: the 2-jet corrects the budget
and can reverse the verdict
010: $\varepsilon_1 = 0.43$, but no flip (curvature > 0)
101: no first-order flip, exact 0.46 (curvature < 0)
χ: $F'' = 0$, first order exact
first order ε_1
second order (2-jet) ε_2
exact
smallest injected share that flips the act
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
record 111
record 001
record 100